Intro To ML
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Introduction

1.1 What is Machine Learning ?

Machine learning = Data →Algorithms →Insight

1.1.1 Data

Data = attributes/tabular, text, speech, images, mixed.

Definition 1 (Data sample)**.** An individual observation. E.g., an image, or a list of attributes for one patient

Definition 2 (Data set)**.** A collection of multiple data sample. E.g., a collection of images, the attribute lists for multiple patients.

Unsupervised data : Each sample consists only of an observation, e.g., an image (without information about its content).
Supervised data : Each sample comes with additional annotations, e.g., category label.

1.1.2 Insight (what we discover from the datas)

What insight ?

- Precise, concrete prediction. E.g., the category depicted by an image.
- Better understanding of a phenomenon. E.g., identify the mother's charasteristics that lead to low birth weight.
- Better understanding of data. E.g., analyze the influences between different musical genres.
- **in this course** we mostly focus on precise insights. In particular we will study two classes of ML problems : **regression** and **classification**

Regression : Predict continuous value(s) for a given sample. E.g., predict the birth weight of a baby, human pose estimation : predict the 3D positions of human joints (multiple values).
Classification : Predict one discrete label for a given sample. E.g., binary classifications for movies (like vs not like; 1 or 0), muti-class image recognition (cat vs tree vs car; 0 vs 1 vs 2).

Order:
In regression, the values typically follow an order. E.g, predicting a weight of 3002g insted of 2977g is better that predicting 2500g.
In classification, the categories do not follow an order. E.g, predicting a "tree" instead of a "cat" is just as wrong as predicting "car". →here it's true or false.

1.1.3 Algorithms

What (classes of) algorithms ?

- Supervised learning :
  - Relies on supervised data
  - The annotations typically correspond to the desired insight
- Unsupervised learning :
  - Relies on unsupervised data
  - The goal is rather to analyze the observed data set

Unsupervised learning:

- Single stage: Analyze the whole data or transform it for further analysis.

Supervised learning:

- Stage 1: Training: Use data with ground-truth (info that is true) labels to optimize model parameters.
- Stage 2: Testing: Predict the output for a new data sample.
- Assumption : The annotations (correct responses) are the insight that we seek to obtain from the data. E.g.,: category label, human pose

Training set vs test set:

- The training set and the test set should always be **completely separate**.
- Never use the test annotations during training →the test set is not to train or to improve the performance of the model, but only to test his efficiency.
- Assumption: they are both drawn from the same statistical distribution. E.g., digit recognition: A model trained on MNIST will work poorly on Street View House Numbers

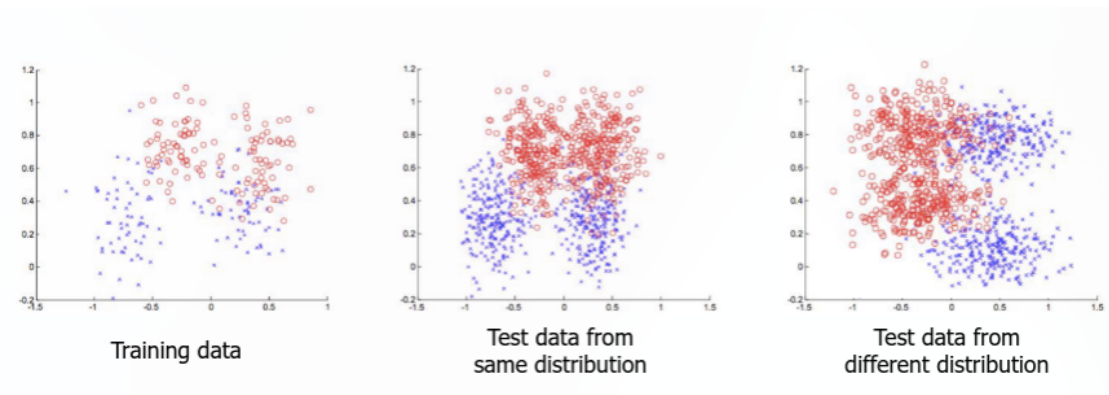


Figure 1.1: Same distribution assumption

Supervised Linear Model

2.1 Notation

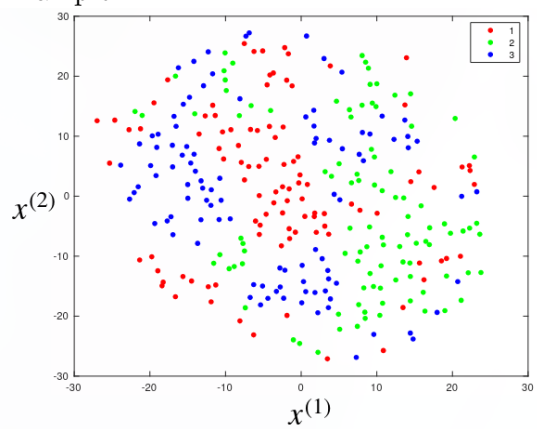
We denote the ith data sample (input) in the collection of N samples as

$$x_i \in \mathbb{R}^D,$$

a column vector of dimension D.
We denote the ith label (output) in the collection of N samples as y_i.
y_i :

- for classification represents the category the sample belongs to.
- for regression, it can be a single continuous value, or a column vector of dim C.

Example:



Here, x_i is one 2D point and thus D = 2, y_i is a discrete value indicating the class (color), i.e, 1, 2 or 3.

2.2 Linear Regression

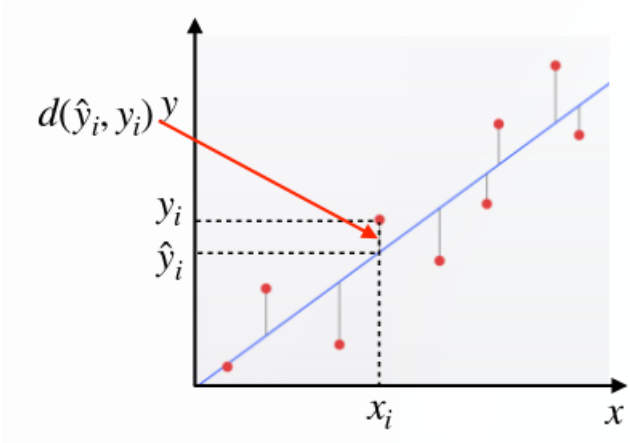
2.2.1 Training

Given N annotated data samples (x_i, y_i) with measurements, we can plot the data, and we will approximate it with a line. Our goal is to find the line that bests fits these measurements →linear regression. It consists of finding the best line parameters $w^{(0)*}$ and $w^{(1)*}$ for some given data.
This corresponds to the training stage:
Given N training pairs (x_i, y_i) , we aim to find $\left(w^{(0)}, w^{(1)}\right)$, s.t the predictions of the model

$$\hat{y}_i = w^{(0)} + w^{(1)} x_i$$

are close to the true values y_i.
The mesure of closeness/error between y_i and $\hat{y}_i$ is represented by the Euclidean distance:

$$d\left(\hat{y}_i, y_i\right) = \sqrt{\left(\hat{y}_i - y_i\right)^2}$$



$\hat{y}_i - y_i$: often reffered to as the *residual*

In practice, one often prefers using the squared Euclidian distance

$$d^2\left(\hat{y}_i, y_i\right) = \left(\hat{y}_i - y_i\right)^2$$

Training can then be expressed as the least-squared minization problem

$$min_{w^{(0)},w^{(1)}} \frac{1}{N} \sum_{i=1}^{N} d^2(\hat{y}_i, y_i)$$

2.2.2 Prediction

Once the best line for the given N observations is found, we can use it to predict a y value for a new x.
Let $w^{(0)*}$ and $w^{(1)*}$ be the best line parameters given the observations.
Then

$$\hat{y} = w^{(0)*} + w^{(1)*}x$$

is the estimate value of y for any x.